

PA1 – Proof by induction

(solutions)

1. When proving a statement $S(n)$ by induction, the base step (basis) always consider $n=0$ or $n=1$ (1 Ponto)

☐ TRUE

☒ FALSE

2. In proof by induction, the proof for the base case(s) is not optional. (1 Ponto)

☒ TRUE

☐ FALSE

3. In proof by induction, we always need to prove a large number of specific cases  (1 Ponto)

☐ TRUE

☒ FALSE

4. A proof by induction only allows us to prove theorems with equalities.  (1 Ponto)

☐ TRUE

☒ FALSE

5. When proving by induction a statement $S(n)$ that holds for every natural number $n \geq 0$:  (1 Ponto)

☒ It is correct to consider for basis $n=0$ and then prove $S(k+1)$ considering the statement $S(k)$ true

☐ It is correct to consider for basis $n=1$ and then prove $S(k+1)$ considering the statement $S(k)$ true

6. There are statements with more than one induction variable (e.g., two or more natural numbers) that can be proved by induction 

(1 Ponto)

☒ TRUE

☐ FALSE

7. When proving by induction a statement $S(n)$, true for every natural number $n \geq 0$, one can use as basis a large natural number and then prove $S(k-1)$ assuming $S(k)$ is true. 

(1 Ponto)

☐ TRUE

☒ FALSE

8. Can we prove by induction that $2^n \geq n+5$ for $n \geq 3$? 

(1 Ponto)

☒ YES

☐ NO

9. Can we prove by induction the statement: if $x \geq 4$, then $2^x \geq x^2$?

(1 Ponto)

☐ NO

☒ YES

☐ Depends if x represents natural numbers or real numbers

10. In a complete binary tree of height h (not considering the leaves), the number of internal nodes (sum of all nodes in the tree except for the leaves) of that tree equals $2h-1$.

Consider we want to prove the truthfulness of the above statement. Which of the following options applies? 

(1 Ponto)

☐ To prove the statement we need to use the "inventor's paradox".

☒ We can prove the statement to be true with an induction proof, using $h=0$ as the base case, assuming the hypothesis valid for h , and proving for $h+1$.

☐ We cannot prove the statement to be true (of false).

☐ We cannot prove the statement to be true using an induction proof, as as such we can state that it is false.